

CLASE 1

September 9, 2026

0.0.1 CLASE 1-METODO DE ELEMENTOS FINITOS

```
[10]: clc
      clear
```

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0.0.2 VARIABLES

Visualizacion de variables

```
[11]: a=1 %los valores se visualizan en la terminal
      b=2; %Los valores no se visualizan en la terminal
```

```
a = 1
```

```
[12]: format short
      pi_aprx1=pi % aproximación con 4 cifras significativas

      format long
      pi_aprx2=pi %aproximacion 16 cifras significativas
```

```
pi_aprx1 = 3.1416
pi_aprx2 = 3.141592653589793
```

```
[14]: who
      whos
```

Variables visible from the current scope:

```
a          b          pi_aprx1  pi_aprx2
```

Variables visible from the current scope:

variables in scope: top scope

Attr	Name	Size	Bytes	Class
====	====	====	=====	=====
	a	1x1	8	double
	b	1x1	8	double
	pi_aprx1	1x1	8	double

pi_aprx2 1x1 8 double

Total is 4 elements using 32 bytes

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0.0.3 OPERACIONES BASICAS

```
[16]: format short
a=3.45
c=4.60

%Se define las operaciones basicas

c=a+b
d=a-b
e=a*b
f=a/b
g=a^b
```

a = 3.4500
c = 4.6000
c = 5.4500
d = 1.4500
e = 6.9000
f = 1.7250
g = 11.903

Calcular la operacion:

$$(1 + 2^4)^2 \cdot \frac{7^3}{4^2 - 1}$$

```
[20]: A=((1+2^2)^2)*7^3/(4^2-1);
disp("El resultado de A es: "),disp(A)
```

El resultado de A es:
571.67

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0.0.4 DEFINICION DE VECTORES

Vectores fila V_{FILA} Vectores Columna $V_{COLUMNA}$

```
[22]: %Vector fila
v_fila=[1 2 3 4 5]
%vector columna
v_columna=[1;2;3;4;5]
```

v_fila =

1 2 3 4 5

v_columna =

1

2

3

4

5